

5.4 Conics Summary Worksheet

1. Find the vertex of the graph $y = 3x^2 - 12x + 7$.

Let's first get our quadratic into vertex form by completing the square:

$$y = 3(x^2 - 4x + \quad) + 7$$

$$y = 3(x^2 - 4x + 4) + 7 - 12$$

$$y = 3(x - 2)^2 - 5$$

So $h = 2$, $k = -5$

Vertex: $(h, k) \rightarrow (2, -5)$

2. Consider the equation of a conic: $x^2 + 2x + y^2 + 8y - 8 = 0$.

- a. Does this equation represent a parabola, ellipse, circle, or hyperbola?

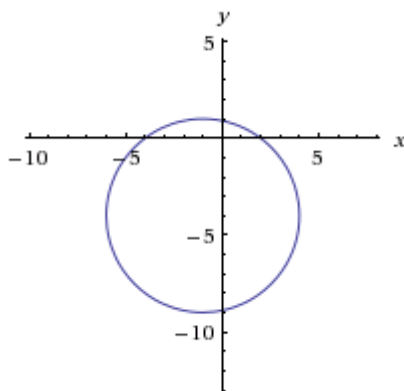
$$(x^2 + 2x + \quad) + (y^2 + 8y + \quad) = 8$$

$$(x^2 + 2x + 1) + (y^2 + 8y + 16) = 8 + 1 + 16$$

$$(x + 1)^2 + (y + 4)^2 = 25$$

So we see that this is the equation of a circle.

- b. Carefully sketch a graph of the above equation.



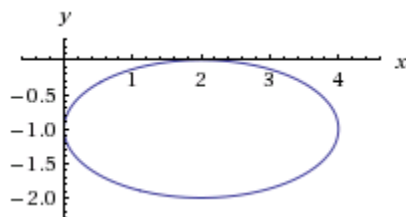
3. Consider the equation of a conic: $-x^2 - 8y - 4 = 4y^2 - 4x$

a. Does this equation represent a parabola, ellipse, circle, or hyperbola?

$$\begin{aligned}
 x^2 - 4x + 4y^2 + 8y &= 4 \\
 (x^2 - 4x + \quad) + (4y^2 + 8y + \quad) &= -4 \\
 (x^2 - 4x + \quad) + 4(y^2 + 2y + \quad) &= -4 \\
 (x^2 - 4x + 4) + 4(y^2 + 2y + 1) &= -4 + 4 + 4 \\
 (x - 2)^2 + 4(y + 1)^2 &= 4 \\
 \frac{(x - 2)^2}{4} + \frac{(y + 1)^2}{1} &= 1
 \end{aligned}$$

So we see that this is the equation of an ellipse.

b. Carefully sketch a graph of the above equation.



4. Consider the equation of a conic: $x^2 - 4y^2 - 2x + 8y - 7 = 0$

a. Does this equation represent a parabola, ellipse, circle, or hyperbola?

$$\begin{aligned}
 x^2 - 2x - 4y^2 + 8y &= 7 \\
 (x^2 - 2x + \quad) + (-4y^2 + 8y + \quad) &= 7 \\
 (x^2 - 2x + \quad) - 4(y^2 - 2y + \quad) &= 7 \\
 (x^2 - 2x + 1) - 4(y^2 - 2y + 1) &= 7 - 4 + 1 \\
 (x - 1)^2 - 4(y - 1)^2 &= 4 \\
 \frac{(x - 1)^2}{4} - \frac{(y - 1)^2}{1} &= 1
 \end{aligned}$$

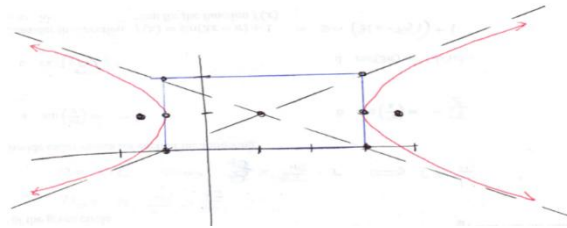
So we see that this is the equation of a hyperbola.

- b. Carefully sketch a graph of the above equation.

Our fundamental rectangle has corners of $(-1,0)$ $(-1,2)$ $(3,0)$ $(3,2)$

Our asymptotes are given by the equations:

$$(y-1) = \pm \frac{b}{a}(x-1) \rightarrow y = \pm \frac{1}{2}(x-1) + 1 \rightarrow \begin{aligned} y &= \frac{1}{2}x + \frac{1}{2} \\ y &= -\frac{1}{2}x + \frac{3}{2} \end{aligned}$$



5. Consider the equation of a conic: $x^2 = 6y - y^2$

- a. Does this equation represent a parabola, ellipse, circle, or hyperbola?

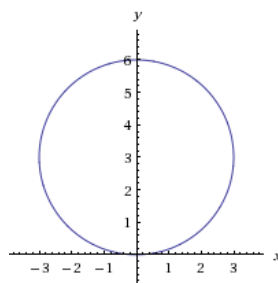
$$x^2 + y^2 - 6y = 0$$

$$x^2 + (y^2 - 6y + 9) = 0 + 9$$

$$x^2 + (y - 3)^2 = 9$$

So we see that this is the equation of a circle.

- b. Carefully sketch a graph of the above equation.



6. Consider the equation: $y^2 - 8x - 8y - 8 = 0$

a. Does this equation represent a parabola, ellipse, or hyperbola?

We can see that there is no x^2 term in this equation, so the equation must represent a parabola that opens left or right.

$$y^2 - 8x - 8y - 8 = 0$$

$$(y^2 - 8y + \quad) - 8x = 8$$

$$(y^2 - 8y + 16) - 8x = 8 + 16$$

$$(y - 4)^2 - 8x = 24$$

$$8x = (y - 4)^2 - 24$$

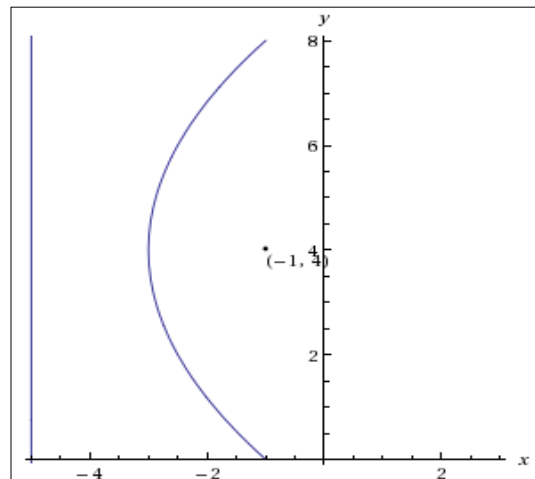
$$x = \frac{1}{8}(y - 4)^2 - 3$$

So we see that this is the equation of a parabola.

From this equation we can also identify the following values:

$$h = -3, \quad k = 4$$

b. Carefully sketch a graph of the above equation.



7. Consider the conic given by the equation: $-x^2 + 2x + 4y^2 + 8y - 8 = 0$.

a. Does this equation represent a parabola, ellipse, or hyperbola?

$$\begin{aligned}
 -x^2 + 2x + 4y^2 + 8y - 8 &= 0 \\
 (-x^2 + 2x + \quad) + (4y^2 + 8y + \quad) &= 8 \\
 -(x^2 - 2x + \quad) + 4(y^2 + 2y + \quad) &= 8 \\
 -(x^2 - 2x + 1) + 4(y^2 + 2y + 1) &= 8 - 1 + 4 \\
 -(x-1)^2 + 4(y+1)^2 &= 11 \\
 -\frac{(x-1)^2}{11} + \frac{4(y+1)^2}{11} &= 1 \\
 \frac{4(y+1)^2}{11} - \frac{(x-1)^2}{11} &= 1
 \end{aligned}$$

So we see that this is the equation of a hyperbola.

b. Carefully sketch a graph of the above equation.

Our fundamental rectangle has corners of

$$\begin{aligned}
 &\left(1 + \sqrt{\frac{11}{4}}, -1 + \sqrt{11}\right) \quad \left(1 + \sqrt{\frac{11}{4}}, -1 - \sqrt{11}\right) \\
 &\left(1 - \sqrt{\frac{11}{4}}, -1 + \sqrt{11}\right) \quad \left(1 - \sqrt{\frac{11}{4}}, -1 - \sqrt{11}\right)
 \end{aligned}$$

Our asymptotes are given by the equations:

$$(y+1) = \pm \frac{b}{a}(x-1) \rightarrow y = \pm \frac{\sqrt{11}}{\sqrt{11/4}}(x-1) - 1 \rightarrow \begin{aligned} y &= 2x - 3 \\ y &= -2x + 1 \end{aligned}$$

